

LÉA

DESSE

EMBEDDED SOFTWARE ENGINEER



Passionate embedded systems engineer with 5+ years of experience in firmware development. Expertise in communication protocols (CAN, I2C, UART, Zigbee, OpenThread) and test automation. Creative profile with several successful personal projects combining hardware and software.

INFORMATIONS

✉ lea.desse@hotmail.fr

📞 06.05.44.77.79

🏠 Mougins

👤 Française

INTERESTS

Electronics
Track motorcycle racing
3D printing
Hiking & camping

Active contributor to
workplace community life

PROFESSIONAL EXPERIENCE

Embedded Software Engineer - STMicroelectronics 2021 - 2025

- Development and maintenance of automated validation tools for **Zigbee** and **OpenThread** stacks
- Continuous integration with **Jenkins** and test process optimization
- Automation of **power consumption** measurements (Joulescope, ST-Link V3 PWR)

Embedded Software Engineer - Elsys Design 2020 - 2021 (contractor at STMicroelectronics)

- Development of embedded tests in **C** for OpenThread integration validation on **STM32WB**
- Development of test automation tools for Zigbee and OpenThread stacks in **Python**

Embedded Software Intern - Elsys Design 2019

- Collection and processing of embedded sensor data (IMU, GPS, radar) on **STM32H7**
- Development of test bench simulating vehicle behavior (**I2C, UART, CAN FD** protocols)

PERSONAL TECHNICAL PROJECTS

Motorcycle Stabilizer Wing System • STM32F4 • CAN • IMU

Real-time attitude correction system with Bosch inertial measurement unit. Dual-box architecture (control + configuration) communicating via CAN. Complete realization: PCB design, firmware, 3D-printed enclosures. Showcased at motorcycle exhibitions.

Connected Thermal Printer • ESP32 • WiFi • UART

WiFi-connected system with API authentication and Access Point configuration. Complete realization: PCB design, ESP32 firmware, 3D-printed enclosure.

NFC Badge Reader System • STM32L0 • NFC • RS232

Firmware development with RS232 communication to Python server (STM32L0, ST25R200).

TECHNICAL SKILLS

Microcontrollers: STM32 (H7, F4, WB, L0), ESP32

Protocols: CAN, CAN FD, I2C, UART, RS232, WiFi, Zigbee, OpenThread, NFC

Tools: Jenkins, Git, Joulescope, ST-Link V3 PWR

Hardware: PCB design, 3D printing, sensors (IMU, GPS, radar)



EDUCATION

Master's Degree in Embedded Systems Engineering

ESME Sudria - Paris | 2019

Specialization in Embedded Systems • 6-month exchange program in Budapest